



香港科技大學
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

ELEC 2100 Signals and Systems

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Outline Chapter 3 Frequency domain analysis of CT signals



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1 Overview

2 Orthogonal Function Decomposition of Signals

3 Fourier Series Analysis of Periodic Signals

4 Spectrum of Aperiodic Signals—Fourier Transform

5 Fourier Transform of Typical Aperiodic Signals

6 Basic Properties of Fourier Transform

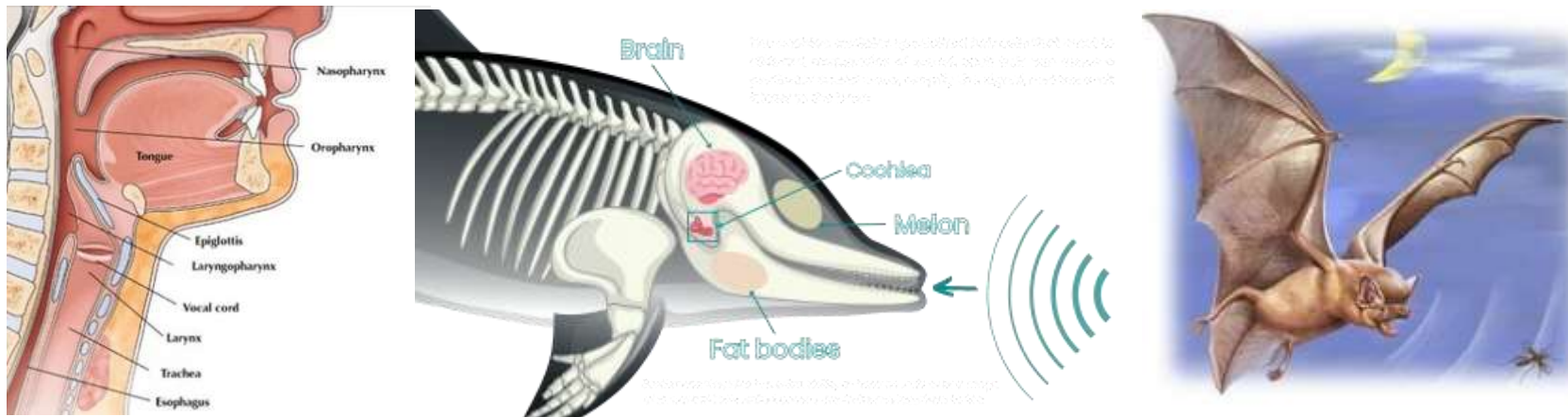
7 Fourier Transform of Periodic Signals

Main Content

1. Learning the concept of spectrum
2. Understanding frequency-domain analysis as a method of decomposing signals into frequency components, analyzing their responses, and reconstructing them

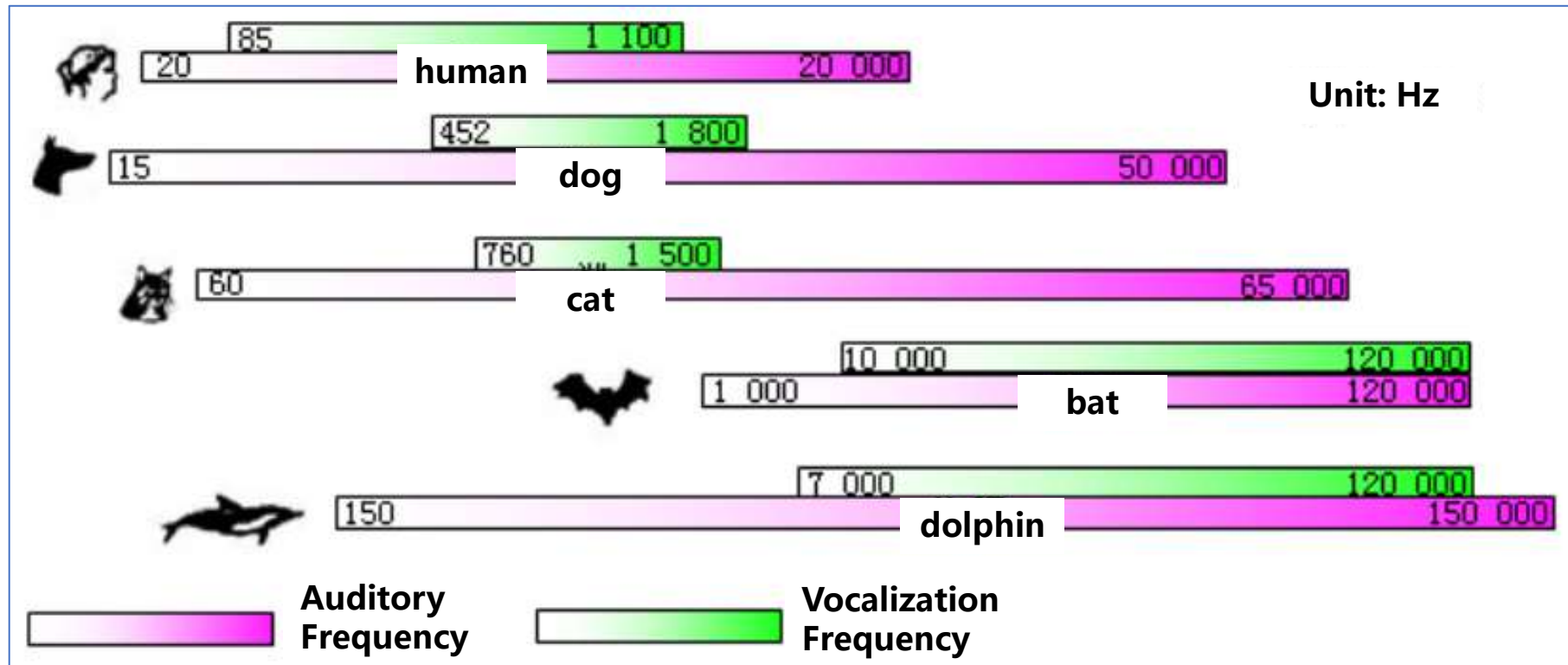
1. Frequency Characteristics and Spectrum

The vocal systems of humans, dolphins, and bats



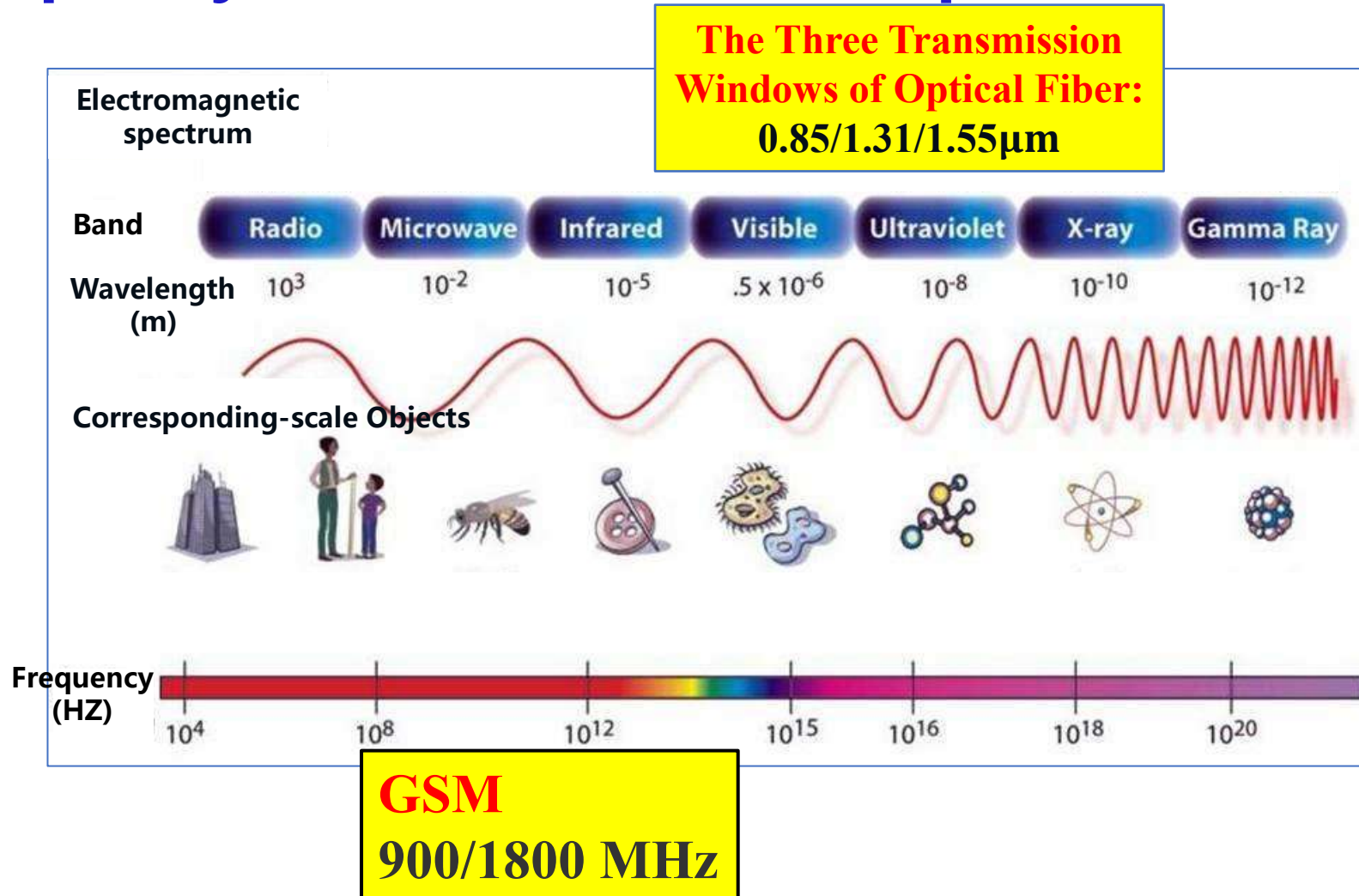
The **frequency characteristics** of signals and systems are determined by their **physical properties**

1. Frequency Characteristics and Spectrum



The **frequency characteristics** of signals and systems are determined by their **physical properties**

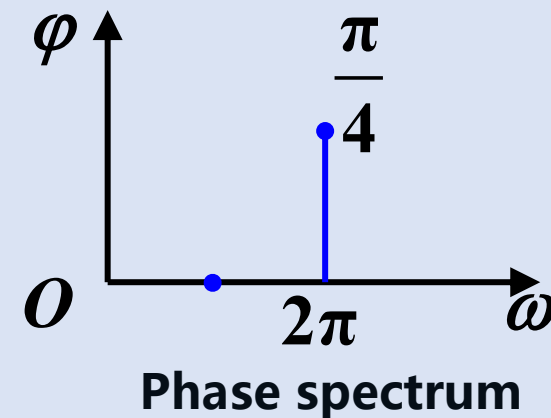
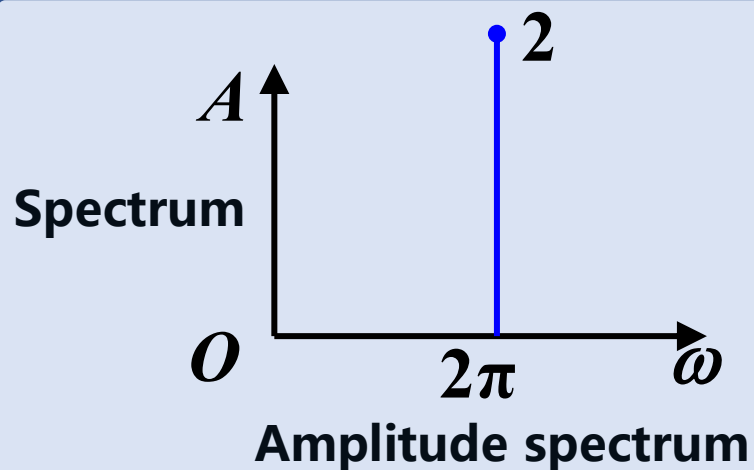
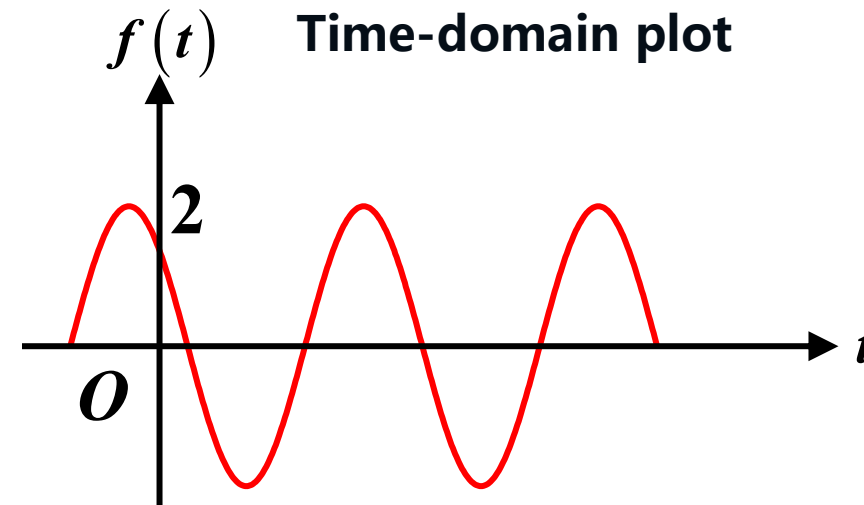
1. Frequency Characteristics and Spectrum



1. Frequency Characteristics and Spectrum

$$f(t) = 2 \cos\left(2\pi t + \frac{\pi}{4}\right)$$

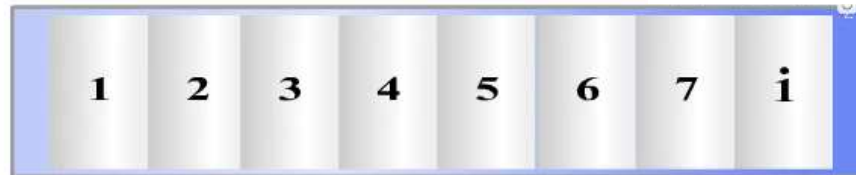
Amplitude $A = 2$
Angular frequency $\omega = 2\pi$
Initial phase $\varphi = \frac{\pi}{4}$



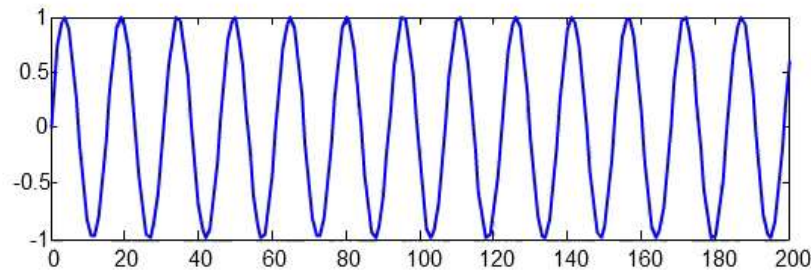
1. Frequency Characteristics and Spectrum

Musical Notation as Spectral Representation

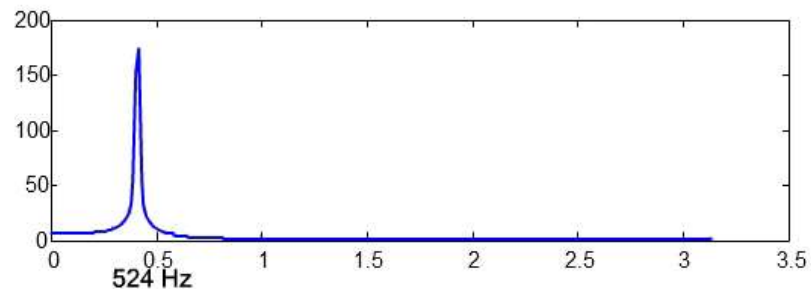
Keyboard



Waveform
(Envelope)



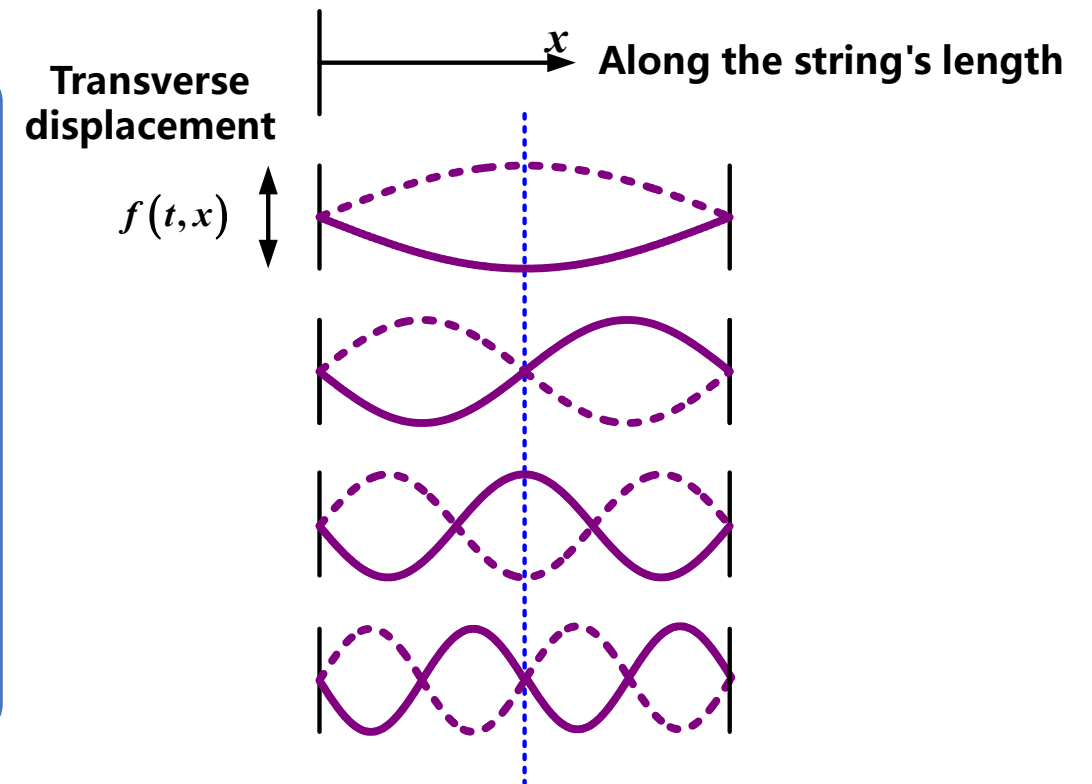
Spectrum



1. Frequency Characteristics and Spectrum

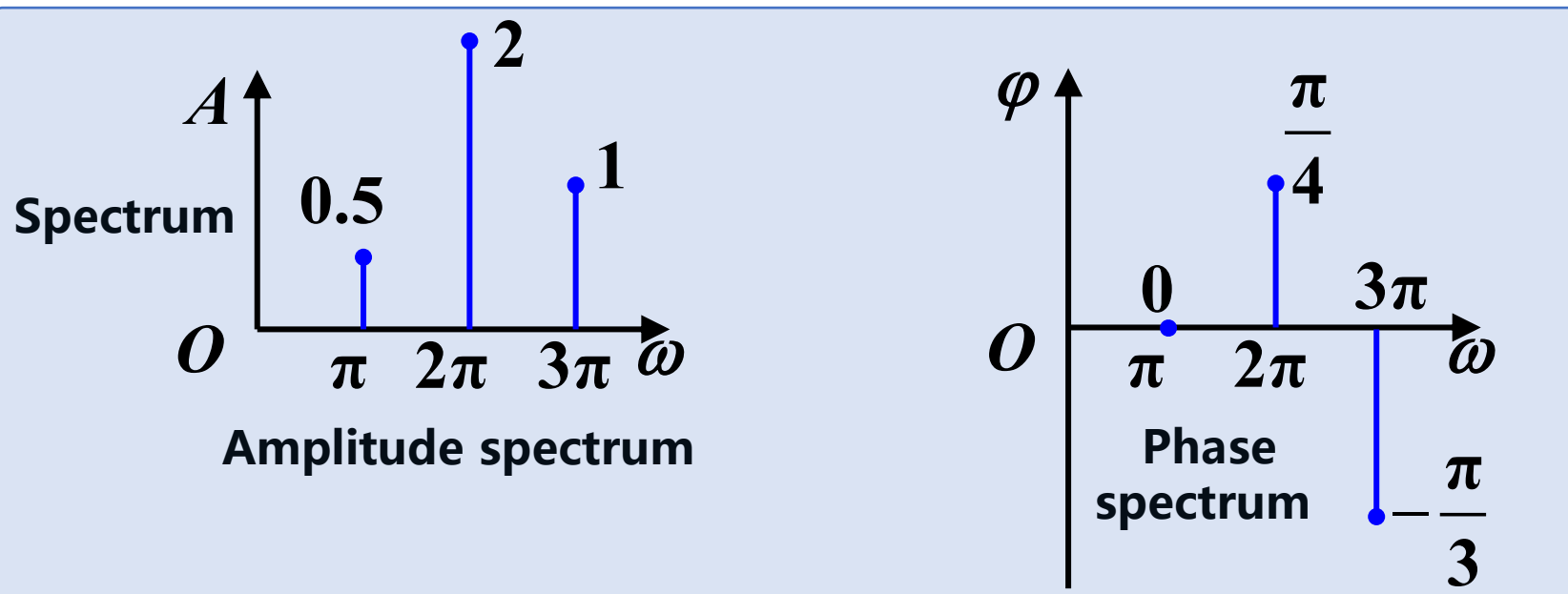
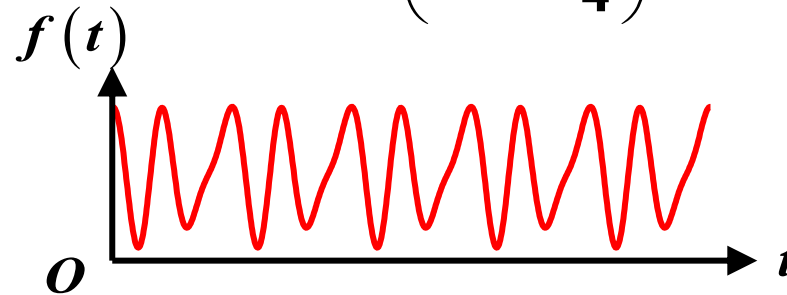
In 1748, Euler, in his study of vibrating strings, discovered that the oscillation modes of strings were all sinusoidal functions of x and exhibited harmonic relationships

- Fundamental (First Harmonic) $\sin(\omega_0 t)$
- Second Harmonic $\sin(2\omega_0 t)$
- Third Harmonic $\sin(3\omega_0 t)$
- Fourth Harmonic $\sin(4\omega_0 t)$
-



1. Frequency Characteristics and Spectrum

$$f(t) = 0.5 \cos(\pi t) + 2 \cos\left(2\pi t + \frac{\pi}{4}\right) + \cos\left(3\pi t - \frac{\pi}{3}\right)$$



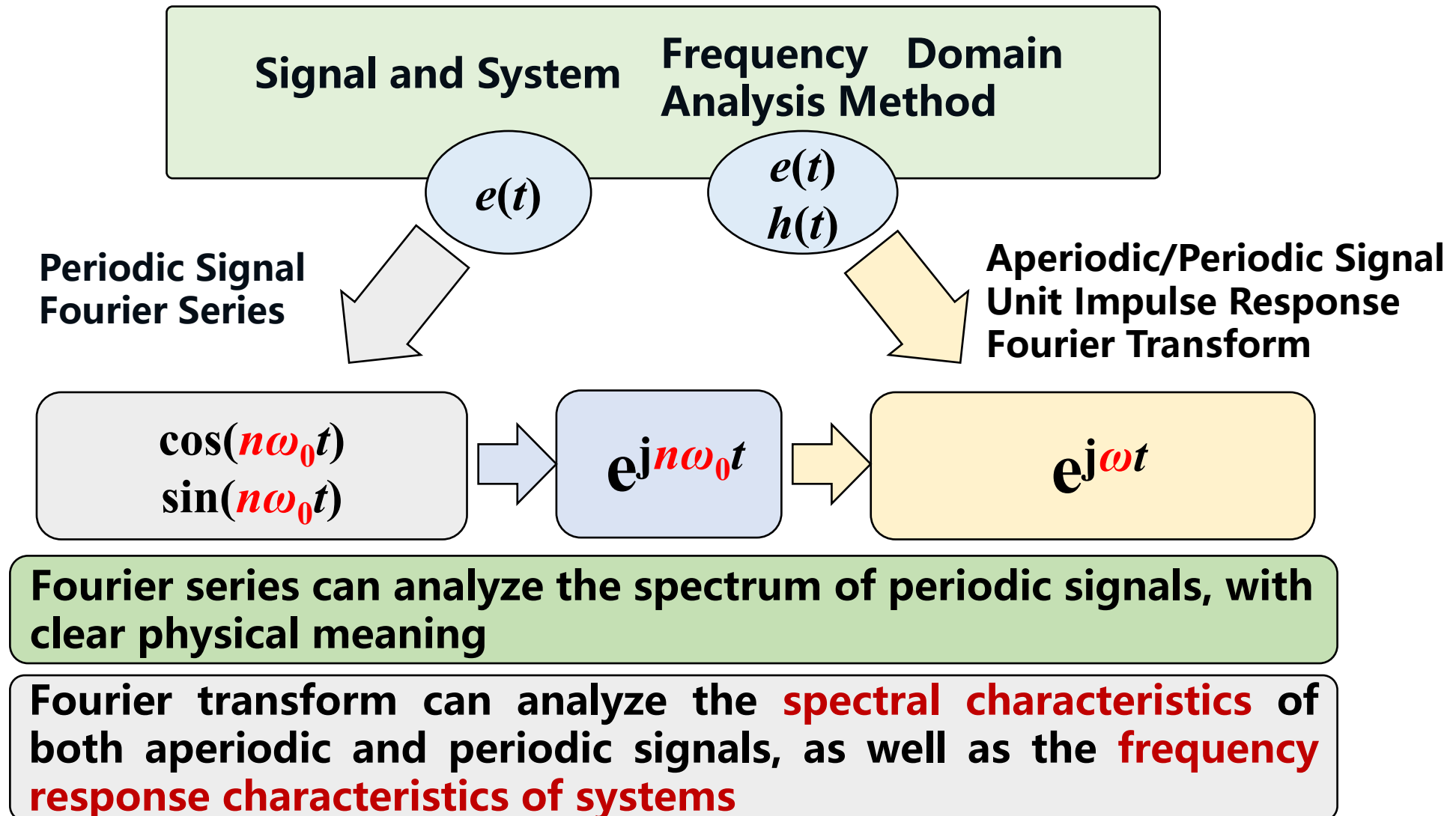
2. Overview of Fourier Frequency Analysis



Periodic signals can all be expanded/decomposed into combinations of their fundamental frequency and harmonic trigonometric components (single-frequency components)

Jean Baptiste Joseph Fourier (1768–1830)
Baron, French mathematician and physicist

2. Overview of Fourier Frequency Analysis



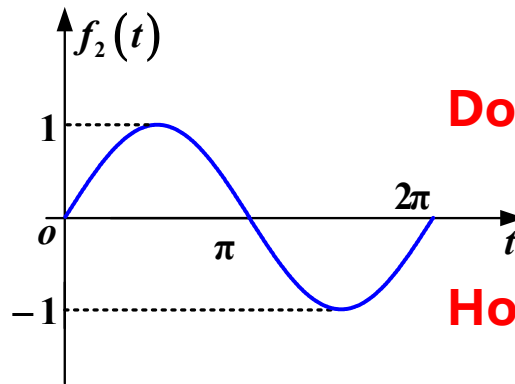
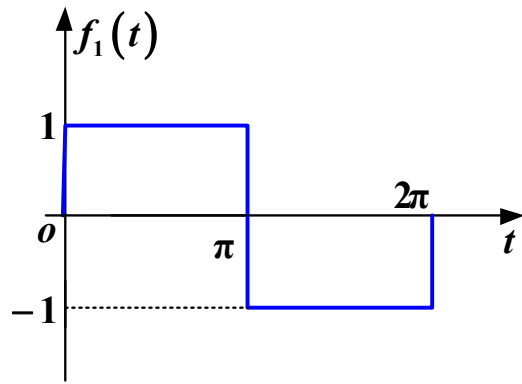
Summary

1. Learning the concept of spectrum
2. Understanding frequency-domain analysis as a method of decomposing signals into frequency components, analyzing their responses, and reconstructing them

Main Content

1. Orthogonal Function Decomposition of Signals
2. Analogy to Vector Orthogonal Decomposition
3. Extraction of Orthogonal Signal Components
4. Real Orthogonal Function Set
5. Complex Orthogonal Function Set
6. Complete Orthogonal Function Set
7. Parseval's Theorem

1. Orthogonal Function Decomposition of Signals



Does $f_1(t)$ contain any $f_2(t)$ components?

How much $f_1(t)$ component is present in $f_2(t)$?

$$f_1(t) = c_{12}f_2(t) + f_e(t)$$

Error Function:

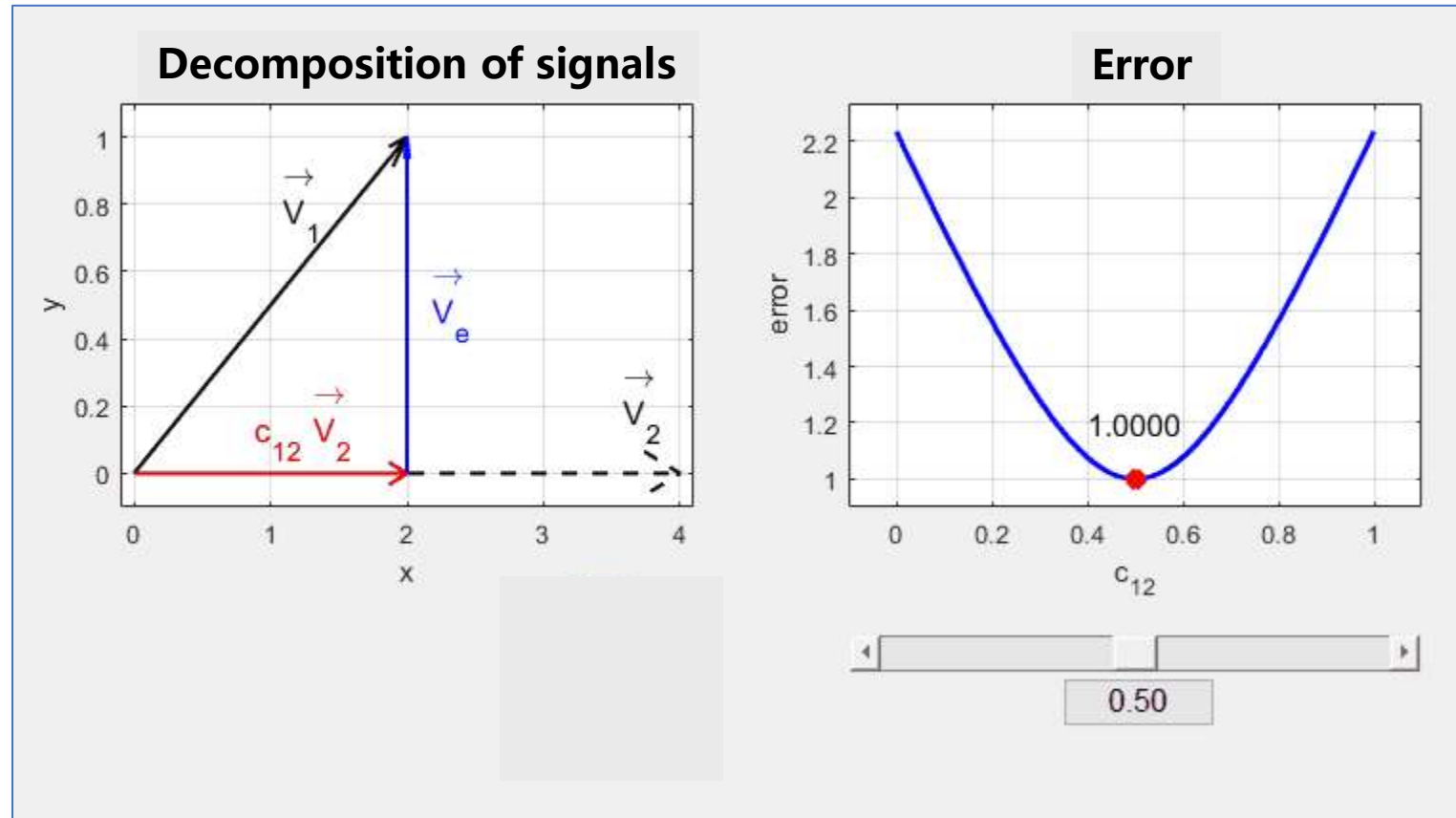
$$f_e(t) = f_1(t) - c_{12}f_2(t)$$

Projection Coefficient: c_{12}

To determine the projection coefficients, the method of **orthogonal function decomposition** must be employed

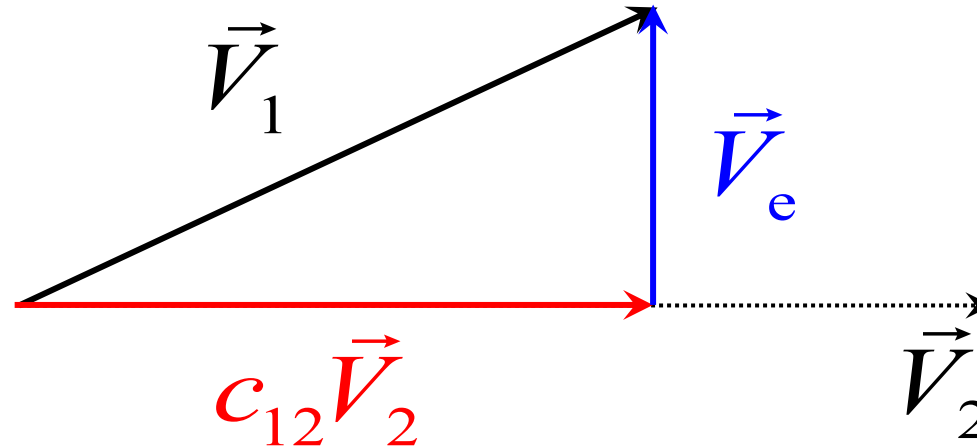
2. Analogy to Vector Orthogonal Decomposition

Example: Given signal $\vec{V}_1$ and component $\vec{V}_2$, the signal can be decomposed into $\vec{V}_1 = c_{12}\vec{V}_2 + \vec{V}_e$. How to select coefficient c_{12} ?



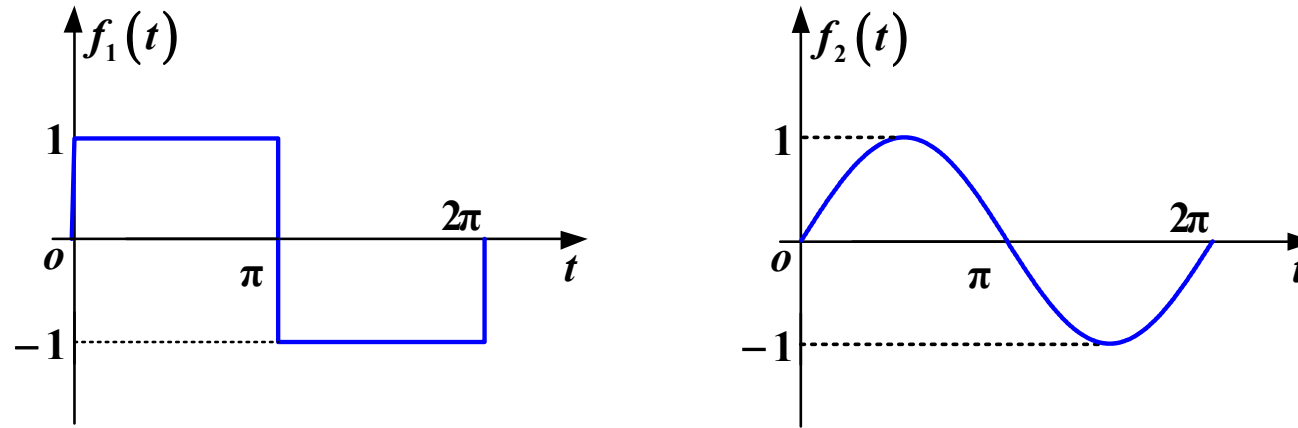
2. Analogy to Vector Orthogonal Decomposition

Selection of Projection Coefficients



- When two vectors are orthogonal, the error component is minimized
- The error component no longer contains any $\vec{V}_2$ component

3. Extraction of Orthogonal Signal Components



$$f_1(t) = c_{12}f_2(t) + f_e(t)$$

Error Function:

$$f_e(t) = f_1(t) - c_{12}f_2(t)$$

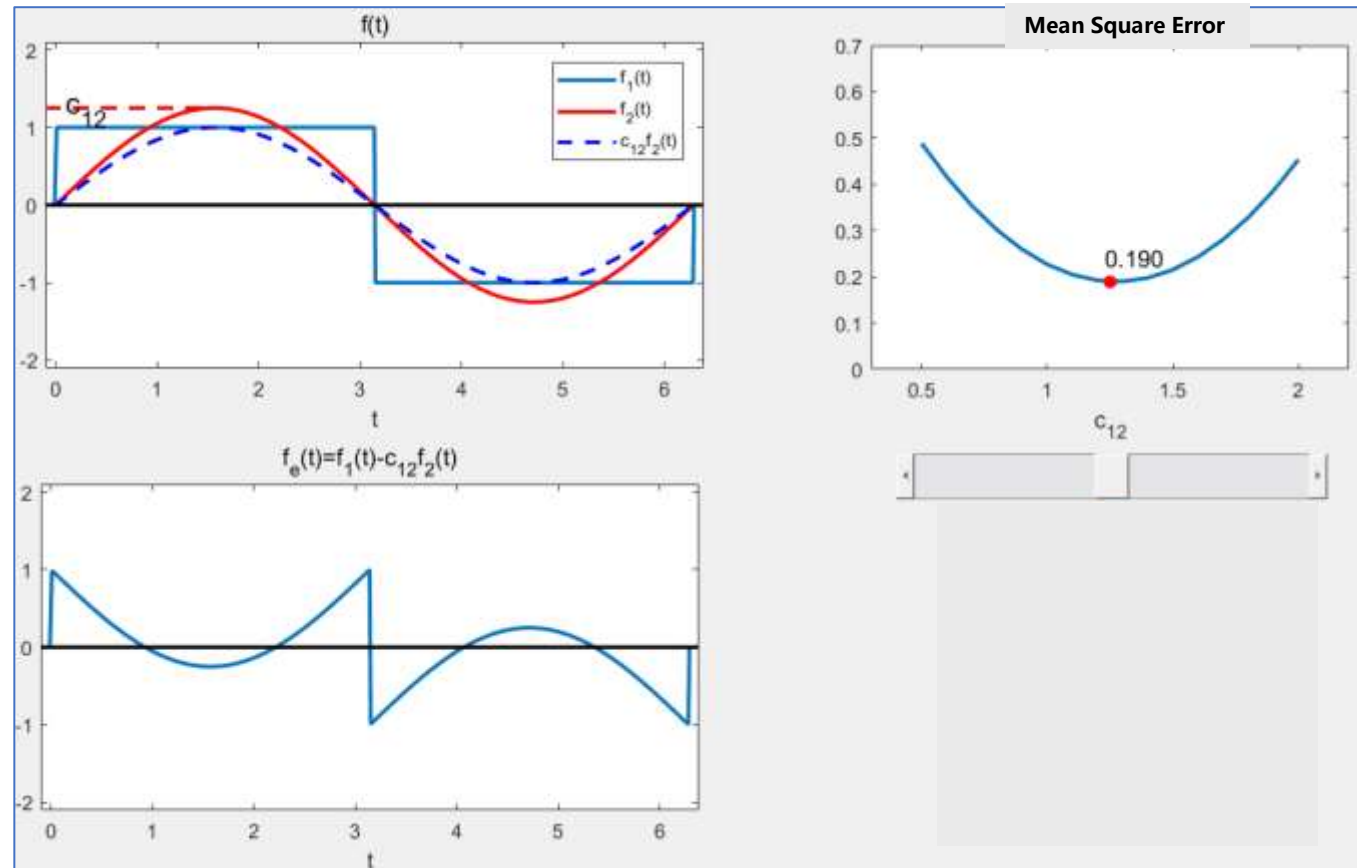
Mean Power (Mean Square Error) of the Error Signal over the Interval (t_1, t_2)

$$\overline{\varepsilon^2} = \overline{f_e^2(t)} = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} [f_1(t) - c_{12}f_2(t)]^2 dt$$

2. Orthogonal Function Decomposition of Signals

3. Extraction of Orthogonal Signal Components

Determination of Projection Coefficients Based on the Minimum Mean Square Error (MMSE) Criterion



3. Extraction of Orthogonal Signal Components

$$\frac{d \overline{\varepsilon^2}}{d c_{12}} = 0 \Rightarrow$$

$$\frac{d}{d c_{12}} \left\{ \int_{t_1}^{t_2} [f_1(t) - c_{12} f_2(t)]^2 dt \right\} = 0$$

Exchange order of differentiation/integration

$$\int_{t_1}^{t_2} \frac{d}{d c_{12}} \left[\underbrace{f_1^2(t)}_{(1)} - \underbrace{2c_{12} f_2(t) f_1(t)}_{(2)} + \underbrace{f_2^2(t) c_{12}^2}_{(3)} \right] dt = 0$$

3. Extraction of Orthogonal Signal Components

$$\int_{t_1}^{t_2} \frac{d}{dc_{12}} \left[\underbrace{f_1^2(t)}_{(1)} - \underbrace{2c_{12}f_2(t)f_1(t)}_{(2)} + \underbrace{f_2^2(t)c_{12}^2}_{(3)} \right] dt = 0$$

$$(1) \frac{d}{dc_{12}} f_1^2(t) = 0 \quad (\text{Because } f_1(t) \text{ does not contain } c_{12})$$

$$(2) \frac{d}{dc_{12}} [-2c_{12}f_1(t) \cdot f_2(t)] = -2f_1(t) \cdot f_2(t)$$

$$(3) \frac{d}{dc_{12}} [c_{12}^2 f_2^2(t)] = 2c_{12} f_2^2(t)$$

3. Extraction of Orthogonal Signal Components

$$\int_{t_1}^{t_2} [-2f_1(t) \cdot f_2(t)] dt + \int_{t_1}^{t_2} 2c_{12} f_2^2(t) dt = 0$$

$$c_{12} = \frac{\int_{t_1}^{t_2} f_1(t) \cdot f_2(t) dt}{\int_{t_1}^{t_2} f_2^2(t) dt}$$

If $c_{12} = 0$, then $f_1(t)$, $f_2(t)$ referred to as orthogonal functions, Thus

$$\int_{t_1}^{t_2} f_1(t) f_2(t) dt = 0$$

4. Real Orthogonal Function Set

$\{g_r(t)\}, r=1, 2, 3 \dots n$
 $g_1(t), g_2(t) \dots g_n(t)$ is called an **orthogonal function set** if

$$\int_{t_1}^{t_2} g_i(t) \cdot g_j(t) dt = \begin{cases} \mathbf{0}, & i \neq j \\ K_i, & i = j \end{cases} \quad (0 \leq i, j \leq n)$$

Definition: All functions in the set must be **pairwise orthogonal**

4. Real Orthogonal Function Set

Any real signal $f(t)$ can be approximated by a sum of n orthogonal functions:

$$\begin{aligned} f(t) &\approx c_1 g_1(t) + c_2 g_2(t) + \cdots + c_n g_n(t) \\ &= \sum_{r=1}^n c_r g_r(t) \end{aligned} \quad (t_1 < t < t_2)$$

$$c_r = \frac{\int_{t_1}^{t_2} f(t) g_r(t) dt}{\int_{t_1}^{t_2} g_r^2(t) dt} = \frac{\int_{t_1}^{t_2} f(t) g_r(t) dt}{K_r}$$

$c_1, c_2, \cdots, c_n$ are independent and can be computed in any order (a property unique to orthogonal functions)

2. Orthogonal Function Decomposition of Signals

Example 3

Prove the trigonometric function set is orthogonal over the interval

$$\left\{ \sin(k\omega_0 t), \cos(k\omega_0 t) \right\}, k = 0, 1, 2, \dots, (0, T_0), T_0 = \frac{2\pi}{\omega_0}$$

$$\sin(n\omega_0 t) \sin(m\omega_0 t)$$

$$= \frac{1}{2} \left\{ -\cos[(n+m)\omega_0 t] + \cos[(n-m)\omega_0 t] \right\}$$

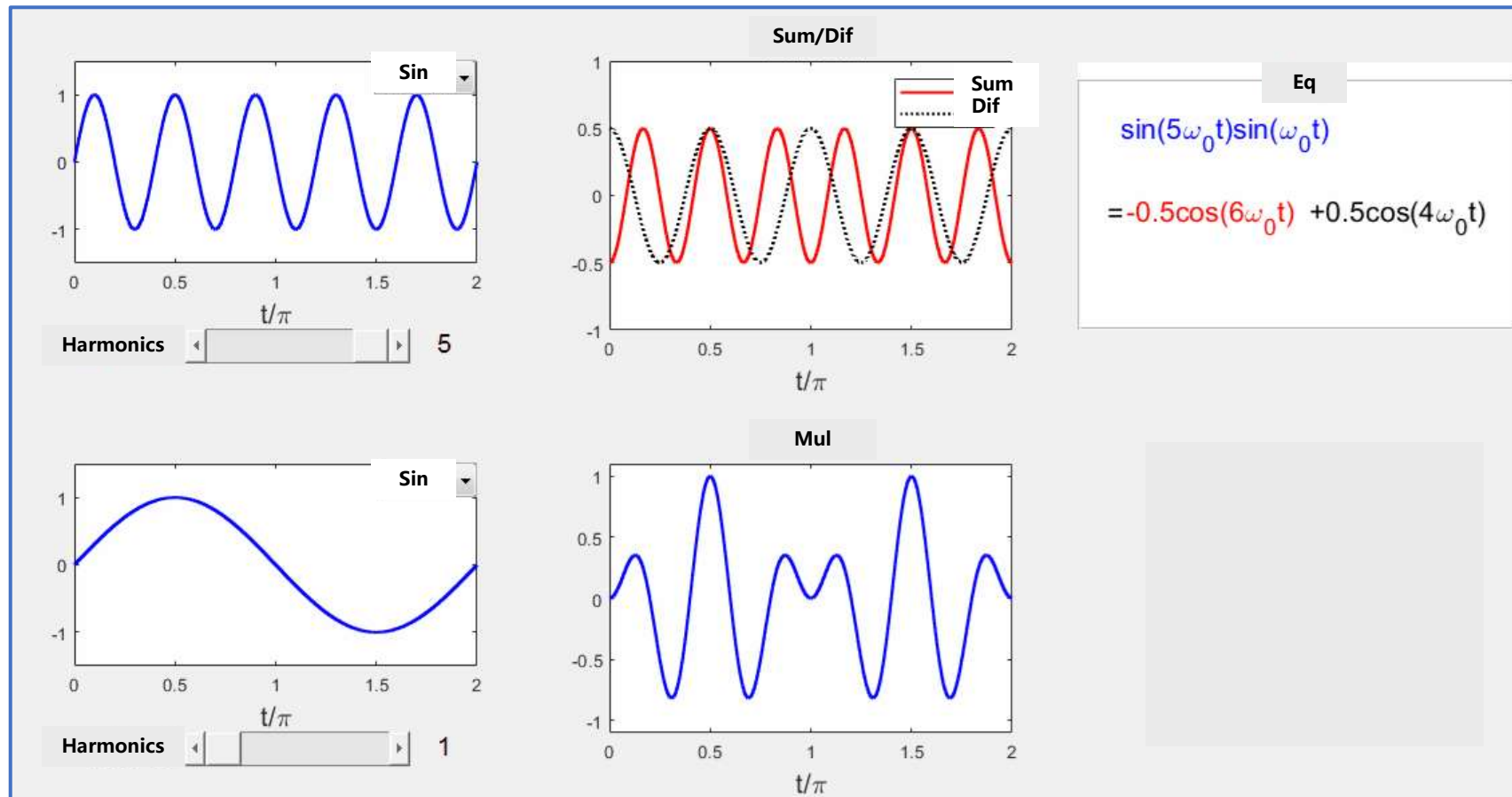
The **sum-frequency term** consists of $n+m$ complete sinusoidal waves over interval $(0, T_0)$

The **difference-frequency term** consists of $n-m$ complete sinusoidal waves over interval $(0, T_0)$

2. Orthogonal Function Decomposition of Signals

Example 3

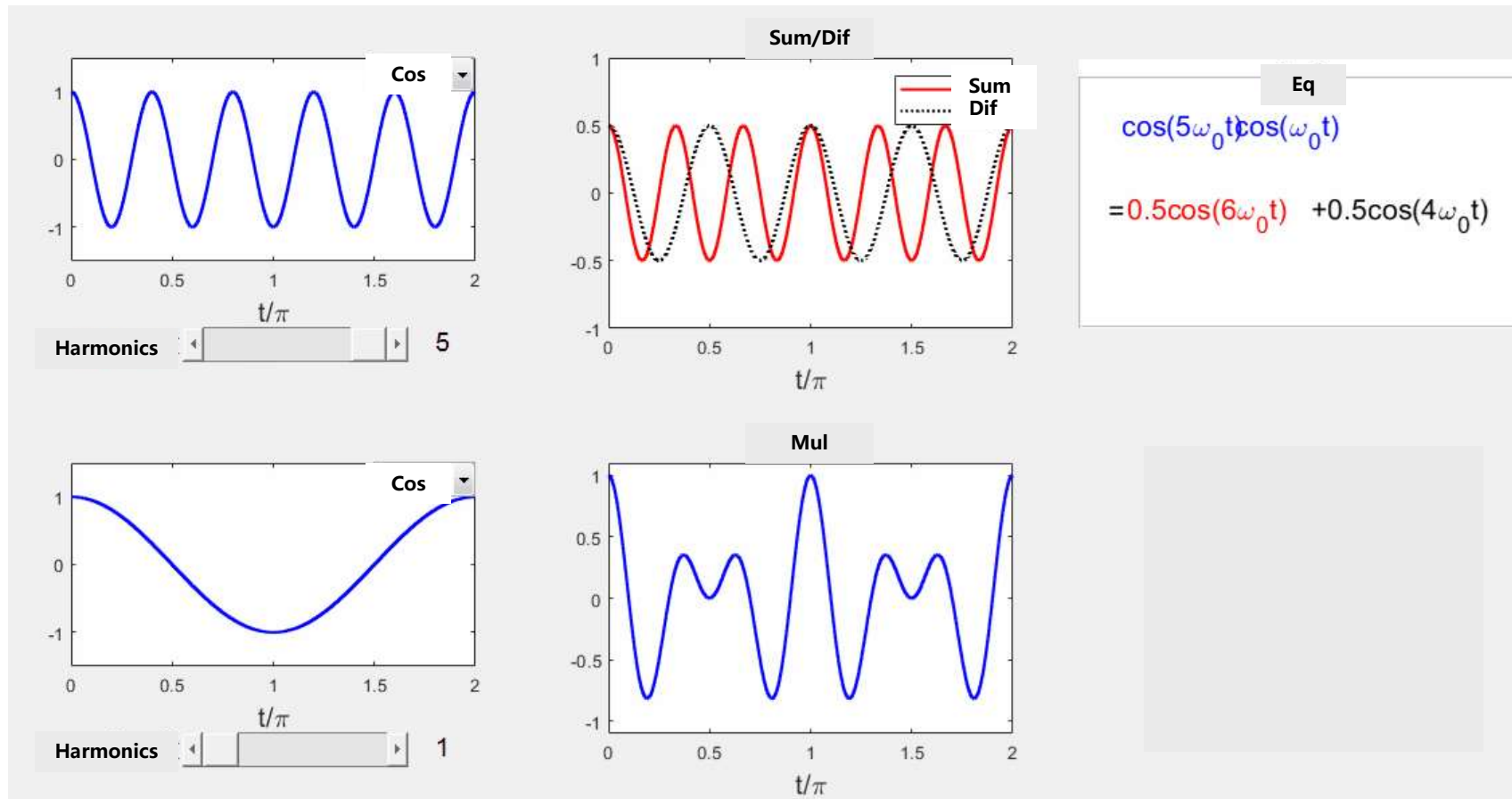
$$\int_0^{T_0} \sin(n\omega_0 t) \sin(m\omega_0 t) dt = \begin{cases} 0 & n \neq m \\ \frac{1}{2}T_0 & n = m \end{cases}$$



2. Orthogonal Function Decomposition of Signals

Example 3

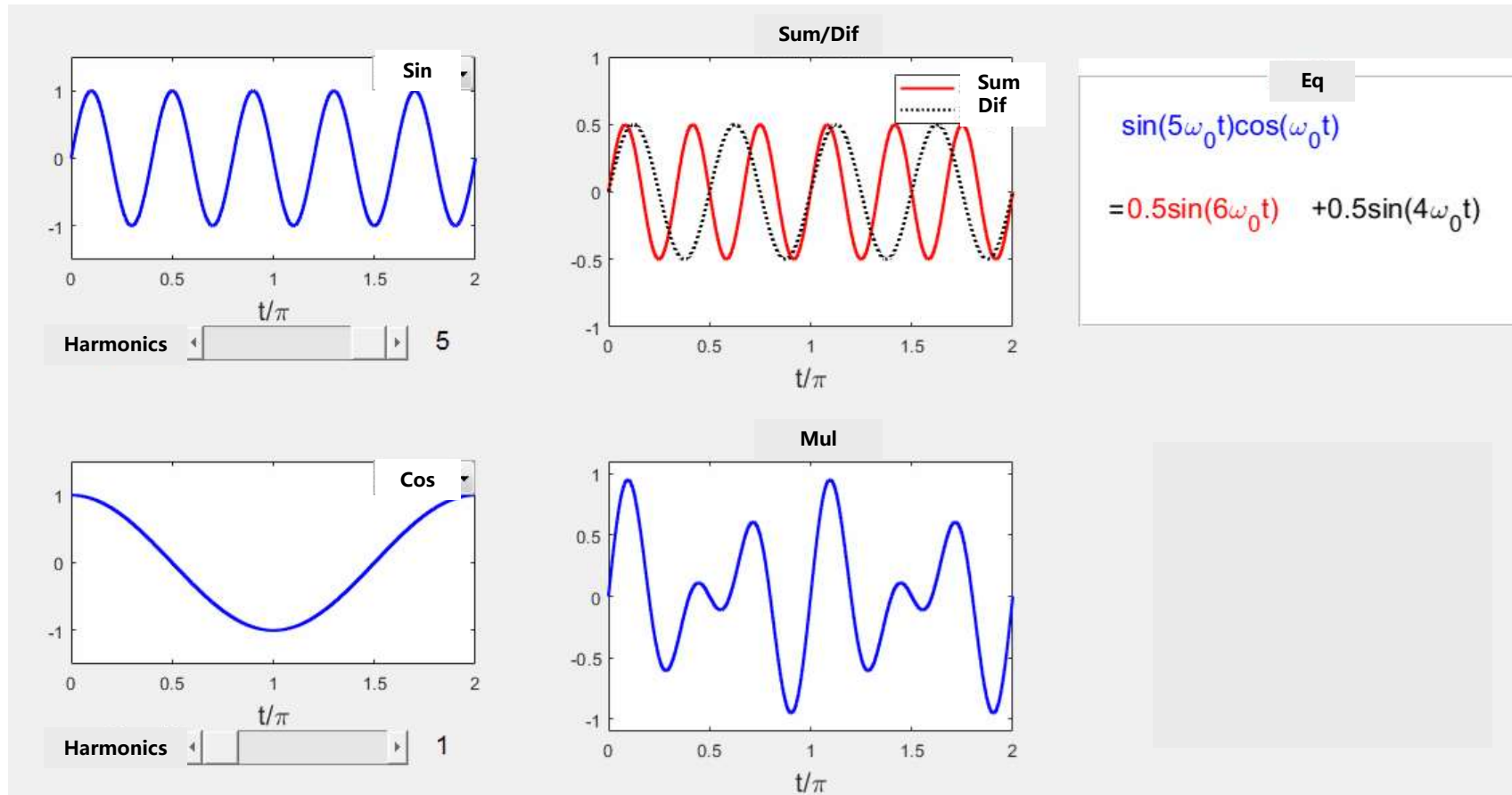
$$\int_0^{T_0} \cos(n\omega_0 t) \cos(m\omega_0 t) dt = \begin{cases} 0 & n \neq m \\ \frac{1}{2} T_0 & n = m \end{cases}$$



2. Orthogonal Function Decomposition of Signals

Example 3

$$\int_0^{T_0} \sin(n\omega_0 t) \cos(m\omega_0 t) dt = 0$$



2. Orthogonal Function Decomposition of Signals

Example 3

$$\int_0^{T_0} \sin(n\omega_0 t) \sin(m\omega_0 t) dt = \begin{cases} 0 & n \neq m \\ \frac{1}{2} T_0 & n = m \end{cases}$$

$$\int_0^{T_0} \cos(n\omega_0 t) \cos(m\omega_0 t) dt = \begin{cases} 0 & n \neq m \\ \frac{1}{2} T_0 & n = m \end{cases}$$

$$\int_0^{T_0} \sin(n\omega_0 t) \cos(m\omega_0 t) dt = 0$$

It satisfies the conditions for an **orthogonal function**, so this function is an orthogonal function set

5. Complex Orthogonal Function Set

A set of complex functions $\{g_r(t)\} (r = 1, 2, \dots, n)$ is **orthogonal** over (t_1, t_2)

$$\text{If: } \int_{t_1}^{t_2} g_i(t) g_j^*(t) dt = \begin{cases} 0 & i \neq j \\ K_i & i = j \end{cases}$$

The signal $f(t)$ can be approximated as:

$$|f(t)|^2 = f(t) \cdot f^*(t)$$

$$f(t) \approx \sum_{r=1}^n c_r g_r(t),$$

$$c_r = \frac{\int_{t_1}^{t_2} f(t) g_r^*(t) dt}{\int_{t_1}^{t_2} g_r(t) g_r^*(t) dt}$$

Example 4

Prove $\{e^{jk\omega_1 t}\}, k = 0, \pm 1, \pm 2 \dots$ over the interval $(0, T_1)$, $T_1 = \frac{2\pi}{\omega_1}$ is an orthogonal function set

Let n, m be any integer, $g_n(t) = e^{jn\omega_1 t}, g_m(t) = e^{jm\omega_1 t}$, Then

$$\begin{aligned} \int_0^{T_1} g_n(t) g_m^*(t) dt &= \int_0^{T_1} e^{jn\omega_1 t} e^{-jm\omega_1 t} dt = \int_0^{T_1} e^{j(n-m)\omega_1 t} dt \\ &= \int_0^{T_1} \cos[(n-m)\omega_1 t] dt + j \int_0^{T_1} \sin[(n-m)\omega_1 t] dt \\ &= \begin{cases} 0 & n \neq m \\ T_1 & n = m \end{cases} \end{aligned}$$

It satisfies the conditions for an **orthogonal function**, so this function is indeed an orthogonal function set.

6. Complete Orthogonal Function Set

$$f(t) \approx \sum_{r=1}^n c_r g_r(t)$$

Definition 1:

As n increases, $\overline{\varepsilon^2}$ decreases. If $n \rightarrow \infty$, then $\overline{\varepsilon^2} \rightarrow 0$. In this case, $g_1(t), g_2(t) \cdots g_r(t) \cdots g_n(t)$ is a **complete** orthogonal function set

Definition 2:

If a function $x(t)$ satisfies $\int_{t_1}^{t_2} g_r(t) \cdot x(t) dt = 0$ for all r , the set is incomplete (unless $x(t)$ is part of the set)

7. Parseval's Theorem

For a signal represented by a complete orthogonal set:

$$f(t) = \sum_{r=1}^{\infty} c_r g_r(t)$$
$$\int_{t_1}^{t_2} |f(t)|^2 dt = \sum_{r=1}^{\infty} \int_{t_1}^{t_2} |c_r g_r(t)|^2 dt$$

Energy of the signal

Energy of each component

The total energy of $f(t)$ equals the sum of energies of its orthogonal components

Proof: Cross-terms vanish due to orthogonality

Summary

Orthogonal Decomposition of Signals

$$f_1(t) = c_{12}f_2(t) + f_e(t)$$

Projection Coefficient

$$c_{12} = \frac{\int_{t_1}^{t_2} f_1(t) \cdot f_2(t) \, dt}{\int_{t_1}^{t_2} f_2^2(t) \, dt}$$

Two signals are orthogonal over the interval (t_1, t_2) if:

$$\int_{t_1}^{t_2} f_1(t) f_2(t) \, dt = 0$$

Summary

- Orthogonal Function Set
- Complete Orthogonal Function Set
- Parseval's Theorem